

BENJAMIN E. JORDAN

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LinkedIn | Personal Website | GitHub

SKILLS

Languages: Python, Java, TypeScript, C++ **Tools:** Git, Docker, AWS, Runpod, Linux, HuggingFace, LaTeX
Libraries: NumPy, PySpark, Pandas, Pytest, PyTorch, OmegaConf, Boto3, Faiss, Optuna, Wandb

EXPERIENCE

Software Engineer @ Amazon Oct 2024 – Present
Measurement, Ad Tech, & Data Science New York, NY

- Productionized and owned team’s brand-awareness model inference system, which processes billions of rows of data daily, producing model predictions and aggregating survey responses for brand lift reports
- Designed and implemented a workflow to automate the training, evaluation, and approval process on a weekly cadence
- Designed a feature to allow shadowing experimental models in a production environment
- Designed and implemented a pre-processing job & data lakehouse for model features which reduced batch inference costs by roughly \$200k per year and enabled the system to scale to new marketplaces
- The system above was referenced as an MLOps template/reference for multiple other large ML projects in the org
- Led design and development of a reporting feature that automatically generates study results in PowerPoint format, saving advertisers hours of manual effort per study

Machine Learning Engineer @ Northrop Grumman Feb 2024 – Oct 2024
Autonomous Intelligence & Robotics Lab Denver, CO

- Enabled accurate, real-time object detection by distilling GroundingDINO into YOLOv10, decreasing inference latency on Jetson Orin from 0.6s to 0.08s
- Researched techniques for implementing and evaluating retrieval-augmented generation (RAG) systems, and led the technical side of a project focused on classified PDF retrieval and question answering
- Implemented a RAG system utilizing ANN search and locally running language models
- Wrote a survey on techniques for mitigating a lack of labeled training data: active learning, semi-supervised learning, knowledge distillation, and synthetic data generation

Machine Learning Engineer @ KLA May 2023 – Aug 2023
Internship Ann Arbor, MI

- Researched and implemented algorithms for semiconductor defect detection and classification
- Fine-tuned and quantized vision transformers with limited data, outperforming statistical methods while staying efficient
- Presented project during poster board session and was invited to give a virtual talk on transformers and transfer learning

PROJECTS

Entropy Audio | Website | Project Blog Jan 2024 – Present
Personal project focused on novel sound design tools for composers

- Curated a multi-terabyte audio sample dataset using web-scraping, open-source data, synthetic data, and manual labeling
- Used local LLMs (Gemma 4), a 5090, and vLLM to efficiently generate captions for millions of samples
- Performed continued pretraining on a 1.4B diffusion transformer after adding new features and text encoders, improved attention masking, additional adaptive layer norms, and improved compatibility with mixed bf16 training
- Implemented policy gradient reinforcement learning algorithms for diffusion models, using pretrained reward models to better optimize for downstream objectives
- Implemented the loss from the Diffusion-DPO paper and trained a model on a synthetic preference dataset
- Designed a frontend that incorporates a preference data flywheel into the user workflow, created a backend service using AWS and RunPod to support generation/inference, as well as other application logic

Open Source Contributions | GitHub April 2026 – Present
• Minor contributions to Google’s JAX, a library for high-performance numerical computing and large-scale machine learning

EDUCATION

Cornell University 2022 – 2023
Master’s in Computer Science, Concentration in Machine Learning Ithaca, NY

Rochester Institute of Technology 2018 – 2022
Bachelor’s in Computer Science, Minor in Music, Presidential Merit Scholarship, Magna Cum Laude Rochester, NY

NOTABLE COURSEWORK

Machine Learning: Large-Scale Machine Learning, Machine Learning Theory, Reinforcement Learning, Machine Learning Hardware and Systems, Computer Vision, Numerical Linear Algebra, Natural Language Processing